EE-170L Computer Programming Lab
The while Loop in C++

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1 Objective

The students will learn and apply the

- The **while** loop
- The nested **while** loop
- The **do...while** loop

2 The while Loop

A **while** loop is a control flow statement that allows code to be executed repeatedly based on a given condition.

The **while** consists of a block of code and a condition.

The condition is first evaluated and if the condition is **true** the code within the block is then executed.

This repeats until the condition becomes **false**.

Syntax:

```
initialization
while          (condition)
{
               statements
               update variable value
}
```

Example:

Write a program using **while** loop who print the values from 5 to 1.

```
#include <iostream>
using namespace std;
int main()
{
    int i = 5; // loop control variable

    while (i > 0)
    {
        cout << i <<"\n";
        i = i - 1;
    }
    cout << "outside loop now";
    return 0;
}
```

Example:

Printing the numbers you entered using a **while** loop. Exit the **while** loop if the number entered is -1.

```
#include <iostream>
using namespace std;
int main()
{
    int num = 5;

    while (num != -1)
    {
        cout << "Enter a number to print: -1 to quit \n";
        cin >> num;
        cout << "You entered: " << num << endl;
    }

    cout << "Outside loop now";
    return 0;
}
```

3 Nested while Loop

A **while** loop inside another **while** loop.

Example:

Write a program that prints the following pattern.

Output:

```
1 2 3 4 5 6 7
1 2 3 4 5 6
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

```
#include <iostream>
using namespace std;
int main()
{
    int num = 7;
    int j;

    while (num > 0)
    {
        j = 1;
        while (j <= num) // nested loop
        {
            cout << j << " ";
            j++; // its equivalent to j=j+1;
        }
        cout << "\n";
        num--;
    }

     return 0;
}
```

4 The do...while Loop

The do . . . while loop is a variant of the while loop with one important difference: the body of do . . . while loop is executed once before the condition is checked.

- The body of the loop executes before checking the condition.
- If the condition is true, the loop body executes again.

- Again the condition is evaluated.
- The process repeats until the condition becomes false.

```
initialization
do {
statements
update variable value
} while (condition);
```

```cpp
#include <iostream>
using namespace std;

int main() {
    int password;

    do {
        cout << "Enter password (1234 to exit): ";
        cin >> password;

        if (password != 1234) {
            cout << "Wrong password! Try again.\n";
        }

    } while (password != 1234);

    cout << "Access Granted!" << endl;

    return 0;
}
```

Lab05: Lab Tasks

1. while Loop

- Write a C++ program that uses a while loop to display numbers from 1 to 10.
- Display each number on a separate line.

2. Repeat the first task using do... while loop

3. Sum using while Loop

- Write a C++ program that prompts the user to enter a positive integer.
- Use a while loop to calculate the sum of all numbers from 1 to the entered integer.
- Display the calculated sum.